

Project Formalization in Lean:

The AKS primality test

–Project Computer Assisted Proofs–

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Prime numbers:

- How do you test whether a given integer n is prime?
 - check for n -divisors $< n$
 - check for n -divisors $\leq \sqrt{n}$
 - check for n -divisors among a subset of integers $\leq \sqrt{n}$
 - Check whether a property of prime numbers holds
- Which primality test is the best?

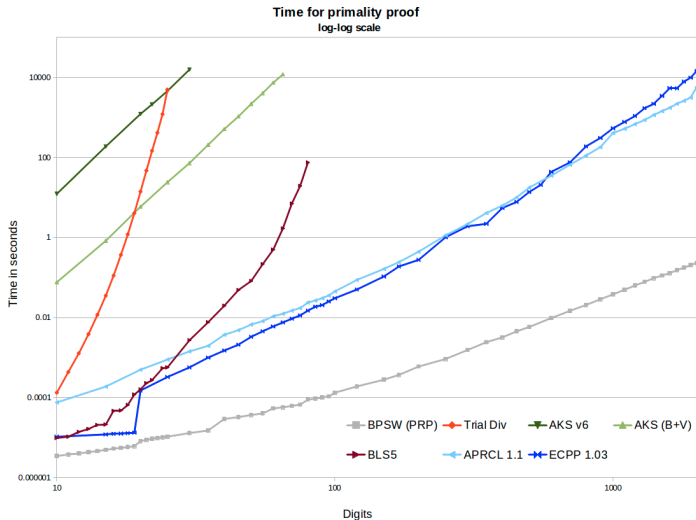
Criteria:

- **Determinism** ensures the algorithm is not probabilistic
- **Independence** of the Riemann Hypothesis
- **Time Complexity**
- **Performance** in practice



Figure: Left-to-right: Saxena, Kayal, Agrawal

- "PRIMES is in P" (2004) [1]
- Claim: AKS is deterministic, independent, and has polynomial time complexity



Theorem (Fermat's little theorem)

p is prime $\implies \forall a \leq p: a^p - a \equiv 0 \pmod p$

- What about the other direction " $\Leftarrow$ "? It doesn't hold for the *Carmichael numbers*: 561, 1729, ...
- Fermat test
- "Using just $a = 2$, only 2183 composite numbers appear prime out of the first 25 billion integers"
- The equivalent in a polynomial ring: $(X + a)^n - (X^n + a) = 0 \pmod n$
- To improve time complexity:

$$(X + a)^n - (X^n + a) = 0 \pmod{(n, X^r + a)}$$

Pseudocode

Input: Integer $n > 1$.

- 1 Check if n is a perfect power: if $n = a^b$ for integers $a > 1$ and $b > 1$, output *composite*.
- 2 Find the smallest r such that $\text{ord}_r(n) > (\log_2 n)^2$. (If r and n are not coprime, then skip this r .)
- 3 For all $2 \leq a \leq \min(r, n-1)$, check that a does not divide n . If $a|n$ for some $2 \leq a \leq \min(r, n-1)$, output *composite*.
- 4 If $n \leq r$, output *prime*.
- 5 For $a = 1$ to $\left\lfloor \sqrt{\varphi(r)} \log_2(n) \right\rfloor$ do
 $\Rightarrow$ if $(X+a)^n \not\equiv X^n + a \pmod{(X^r-1, n)}$, output *composite*.
- 6 Output *prime*.

Let $n = 10001$

- 1 Indeed, 10001 is no perfect power.
- 2 We find $r = 41$.
- 3 For $a = 2, \dots, 41$ we find that 10001 is not divisible by a .
- 4 Clearly, $n \not\leq r$.
- 5 we see that the equality already fails for $a = 1$.

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```
def perfect_pow(n):
    for b in range(2, floor(log(n,2))):
        for a in range(2, floor(sqrt(n))):
            if log(a,10) == log(b,10)*log(n,10):
                return 0
    return n

def find_r(n):
    maxk = floor(log(n,2)^2);
    maxr = max(3, ceil(log(n,2)^5));
    nextR = True;
    k = 1;
    r = 2;
    while nextR == True and r < maxr:
        nextR = False;
        r = r + 1;
        while nextR == False and k <= maxk:
            nextR = (mod(n^k, r) == 1 or
                    mod(n^k, r) == 0)
            k = k+1;
    r = r-1;
    return r;
```

```
def aks_fun(n):
    if perfect_pow(n) == 0:
        return "composite"
    r = find_r(n);
    for a in range(2, min(r+1, n)):
        if mod(a, n) == 0:
            return "composite"
    if n <= r:
        return "prime"
    Z = Integers(n)
    R = PolynomialRing(Z, 'X')
    X = R.0
    X in R
    True
    l = X^r-1;
    l = R.ideal(l)
    R = sage.rings.quotient_ring.QuotientRing(R,
        l, 'X')
    for b in range(
        1, floor(sqrt(euler_phi(r)*log(n,2)))
    ):
        if R((X+b)^n) != X^n+b:
            return "composite"
    return "prime"
```

Theorem (Agrawal, Kayal and Saxena)

Given an integer $n \geq 2$, let $0 \leq r < n$ for which $\text{ord}_r(n) > (\log n)^2 \bmod r$. Then n is prime if and only if

- *n is not a perfect power,*
- *n does not have any prime factor $\leq r$,*
- *$(X + a)^n \equiv X^n + a \bmod (n, X^r - 1) \forall a \in \mathbb{Z}$ such that $1 \leq a \leq \sqrt{r} \log n$.*

- Forward direction: relatively easy
- Backward direction: more difficult

Theorem (Child's Binomial Theorem)

$$n \text{ is prime} \iff (X + a)^n \equiv X^n + a \pmod{n}$$

```
theorem childBinomial (n : ℕ)(a : ZMod n) (two_le_n: 2 ≤ n):
  Nat.Prime n ↔ (X + C a)^n = (X^n + C a)
```

Theorem (Agrawal, Kayal and Saxena)

If n is not a perfect power, and that n does not have any prime factor $\leq r$, then

$$n \text{ is prime} \iff (X + a)^n \equiv (X^n + a) \pmod{(n, X^r - 1)}$$

```
theorem AKS ... (...) ... :
  Nat.Prime n ↔ ∀ a : ℕ, a ≤ (Real.sqrt ↑r) * Real.logb 2 ↑n → (X + C (a : ZMod n))^n - (X^n +
  C (a : ZMod n)) ∈ I
```

$$n \text{ is prime} \implies (X + a)^n \equiv (X^n + a) \pmod{(n, X^r - 1)}$$

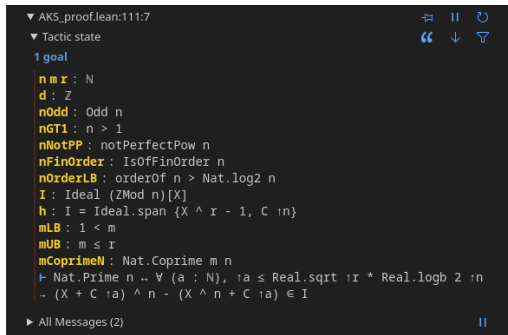
theorem AKS

```
(n m r: ℕ) (d : ℤ)

(nOdd : Odd n) (nGT1 : n > 1) (nNotPP : notPerfectPow n)
(nFinOrder : IsOfFinOrder n)
(nOrderLB : orderOf n > (Nat.log2 n))

(I : Ideal (ZMod n)[X])
(h : I = Ideal.span {X^r - 1, C ↑n})

(mLB : 1 < m) (mUB : m ≤ r) (mCoprimeN : m.Coprime n)
:
  — Nat.Prime n ↔ Ideal.Quotient.mk I (X + C a)^n =
    Ideal.Quotient.mk I (X^n + C a)
Nat.Prime n ↔ ∀ a : ℕ, a ≤ (Real.sqrt ↑r) * Real.logb 2 ↑n →
  (X + C (a : ZMod n))^n - (X^n + C (a : ZMod n)) ∈ I
:= by
```



AKS_proof.lean:111:7

Tactic state

1 goal

```
n m r : ℕ
d : ℤ
nOdd : Odd n
nGT1 : n > 1
nNotPP : notPerfectPow n
nFinOrder : IsOfFinOrder n
nOrderLB : orderOf n > Nat.log2 n
I : Ideal (ZMod n)[X]
h : I = Ideal.span {X ^ r - 1, C ↑n}
mLB : 1 < m
mUB : m ≤ r
mCoprimeN : Nat.Coprime m n
⊢ Nat.Prime n ↔ ∀ (a : ℕ), ↑a ≤ Real.sqrt ↑r * Real.logb 2 ↑n
→ (X + C ↑a) ^ n - (X ^ n + C ↑a) ∈ I
```

► All Messages (2)

$$n \text{ is prime} \implies (X + a)^n - (X^n + a) \in (n, X^r - 1)$$

theorem AKS

```
(n m r: ℕ) (d : ℤ)

(nOdd : Odd n) (nGT1 : n > 1) (nNotPP : notPerfectPow n)
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:= by
```

```
AKS_proof.lean:111:7
Tactic state
1 goal
n m r : ℕ
d : ℤ
nOdd : Odd n
nGT1 : n > 1
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I : Ideal (ZMod n)[X]
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⊢ Nat.Prime n ↔ ∀ (a : ℕ), ↑a ≤ Real.sqrt ↑r * Real.logb 2 ↑n
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▶ All Messages (2)
```

$$n \text{ is prime} \implies (X + a)^n - (X^n + a) \in (n, X^r - 1)$$

```

Nat.Prime n ↔ ∀ a : ℕ, a ≤ (Real.sqrt ↑r) * Real.logb 2 ↑n → (X + C (a : ZMod n))^n - (X^n + C
(a : ZMod n)) ∈ I
:= by
constructor
case mp => -- Nat.Prime n → (X + C ↑a)^n - (X^n + C a) ∈ Ideal.span {X^r - 1, C ↑n}
  rintro hn a ha
  have h1 : (X + C (a : ZMod n))^n - (X^n + C (a : ZMod n)) ∈ Ideal.span {C ↑n} := by
    rw [Ideal.mem_span_singleton]
    have : Fact (Nat.Prime n) := Fact.mk hn
    rw [(childBinomial _ _ nGT1).mp hn, sub_self];
    exact AdjoinRoot.mk_eq_zero.mp rfl

  rw [Ideal.span_insert]
  apply Ideal.mem_sup_right
  exact h1

```

Challenges

- Only one week to get acquainted with Lean
- Certain constructions are obscure to the neophyte
- Lean is very much still in development
- Documentation sometimes abscond and difficult to search
- Doing formal proving is quite different from usual proofs.

Take-aways

- Developed a (moderate) fluency with Lean
- Better understanding of theorem prover
- Better understanding of what it takes to formalize a proof

Summary of our project

- Understand the algorithm and its proof of correctness
- Implemented the AKS algorithm in Sage
- Learn Lean 4
- Formalized the forward direction of Child Binomial Theorem and AKS in Lean, as well as part of the backward direction.

Thank you for your attention !

- [1] M. Agrawal, N. Kayal, and N. Saxena, "Primes is in p ," *Annals of Mathematics*, vol. 160, no. 2, pp. 781–793, 2004.
- [2] A. Granville, "It is easy to determine whether a given integer is prime," *Bulletin of the American Mathematical Society*, vol. 42, September 2004.

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